

**Detroit Transportation Survey: Multi-Stage Area Probability Sample Design: A Technical
Sampling Report**

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Introduction

This report describes the design of a multi-stage area probability sample for a face-to-face survey of Detroit, Michigan residents examining transportation security and use. Key outcome variables include the proportion of residents in households without a vehicle, the proportion whose main means of transportation is public transit, the proportion who skip destinations due to transportation problems, and the proportion who sought transportation help from friends, family, or neighbors in the past month. The sample uses a three-stage design: census tracts as primary sampling units (PSUs), block groups (BGs) as secondary sampling units (SSUs), and persons as elements. Because no address listing or roster of residents exists prior to fieldwork, person-level selection is not executed here. Instead, Section 4 describes the field procedure by which interviewers would list households in each selected BG and select one eligible adult per household — a standard approach in area probability sampling when no person-level frame is available (VDK Ch. 10). For managers, Sections 3.1–3.2 describe the design goals and target population, Section 6 summarizes the weighting plan in non-technical terms, and Section 8 provides maps of the frame and selected sample. For technical personnel, Sections 3.3–4.3 cover frame construction, MOS computation, selection probabilities, and verification; Section 5 addresses anticipated precision; and Section 7 provides variance estimation guidance including how to define strata and PSUs for software implementation. The remainder of this report proceeds as follows: Section 3 covers frame construction and cleaning, MOS computation, and a discussion of alternative sampling rate approaches. Section 4 describes sample selection and the field procedure for person-level sampling. Sections 5 through 7 address precision, weighting, and variance estimation. Section 8 presents maps, and the Appendix provides the codebook, sample listing with domain-specific selection rates, and AI disclosure.

Sample Design

##Design Goals

The sample design for this survey serves to satisfy four requirements simultaneously: obtaining the target of 1,000 total respondents split equally across four age domains (250 per age domain: 18–34, 35–49, 50–64, 65+); a self-weighting sample (a constant f_d per domain) within each domain so that no within-domain weighting adjustments are needed; equal interviewer workload across all 40 sample PSUs to support cost control and field management; and PPS selection at both the PSU and SSU stages. The first three objectives are achieved jointly through the composite measure of size (MOS) technique. (VDK Ch. 10, Sec. 10.5). Selecting one adult per household is also substantively motivated: the key outcome variables are largely household-level characteristics (e.g., vehicle ownership is shared among all household members), so selecting multiple adults from the same household would introduce correlated observations and inflate the design effect.

##Target Population

The target population for this survey are non-institutionalized adults age 18+ who reside in the city of Detroit, MI. This is comprised of approximately 480,131 adults across 270 eligible census tracts according to the 2020 Decennial Census.

##Area Sampling Frame

The data used to construct the sampling frame were sourced from two files : Census Tract FIPS Code Detroit MI.csv: 280 FIPS codes identifying Detroit census tracts, from the City of Detroit Open Data Portal; and Detroit MI.xlsx: tract- and block-group-level population data provided by the course instructors, originating from the 2020 Decennial Census DHC Summary File for Wayne County, MI. It includes housing unit counts, total person counts, and person counts by sex and age from the P12 table (Sex by Age). For the PSU frame, Tract 26163985500 (tract 9855) appears identically 25 times in the data with zero population in every column, meaning it is likely a water body or uninhabited industrial zone. Since it contributes no eligible population, all instances were dropped. The SSU frame consists of 627 block groups, which were reduced to 612

after removing BGs with zero adults. In summary, to clean the frame, zero-population tract 9855 was dropped, along with 5 zero-adult tracts and 9 zero-adult BGs.

##Measures of Size (MOS)

Instead of sizing each tract by its total population, the composite MOS weights each tract's population by how hard each age group is to reach. Groups with lower expected response rates receive higher weight so that tracts are selected with a probability that reflects the actual survey effort they will require across all four age domains. A domain-specific sampling rate

$f_d = n_d^{\text{select}}/N_d$ is defined for each age group, where $n_d^{\text{select}} = 250/r_d$ is the number of persons that must be contacted to yield 250 respondents given expected response rate r_d , and N_d is the total population of that domain in the frame. The composite MOS for tract i is then $S_i = \sum_d f_d N_i(d)$ (VDK Eq. 10.1), a weighted sum of the domain population counts in that tract, and tracts are selected with probability proportional to S_i , giving $\pi_i = mS_i/S$ where $S = \sum_i S_i$ and $m = 40$.

Within each selected tract, one BG is selected with probability proportional to its own BG-level composite MOS $S_{ij} = \sum_d f_d Q_{ij}(d)$, where $Q_{ij}(d)$ is the population of domain d in BG ij , giving a conditional BG selection probability of $\pi_{j|i} = \bar{n} \cdot S_{ij}/S_i$ where $\bar{n} = 1$. Within the selected BG, persons in domain d are then sampled at rate $\pi_{k|ij}(d) = \bar{q} f_d / S_{ij}$, where $\bar{q} = 53.86$ is the target number of persons to contact per BG. The three selection probabilities multiply to

$\pi_i \cdot \pi_{j|i} \cdot \pi_{k|ij}(d) = f_d$ (VDK Sec. 10.5), so that every person in domain d has the same overall probability of selection regardless of which tract or BG they live in, the self-weighting property.

The equal workload property follows from the same construction: summing the expected domain contacts within any BG yields $\sum_d \bar{q} f_d Q_{ij}(d) / S_{ij} = \bar{q}$, so the total number of persons to be contacted is the same in every sample BG and aggregating across the single selected BG per tract, in every sample PSU. The resulting rates for each domain are as follows: $f_{18-34} = 0.00443$, $f_{35-49} = 0.00490$, $f_{50-64} = 0.00425$, $f_{65+} = 0.00438$. The expected workload per BG: $\bar{q} \approx 53.9$ persons to contact.

##Descriptive Statistics on the Frame

After linking, the total Detroit adult population would come up to ~480,131 across 265

PSUs. The Domain population counts would be as follows: 18–34: ~161,357; 35–49: ~113,272; 50–64: ~117,763; 65+: ~87,739 • BG count per tract, distribution of MOS values, and summary of age-domain shares • Identify 5 undersized tracts (all BGs violate feasibility); linked to nearest FIPS neighbor ADDRESS THE ELEPHANT HERE

##Uniform Rate Discussion

One question that comes up during the sample design is: Could a single rate $f = n^{\text{select}}/N_{\text{adult}}$ serve all domains? While an advantage of the method is simpler field implementation, its use would require post-hoc non-response adjustment. When tested, it is found that response rates differ substantially by domain, ranging from 0.35 to 0.65, demonstrating that a uniform rate under-samples low-response domains and therefore should not be used for all. In particular, the 35–49 domain falls ~8.5% short of the 250 person target under uniform rate.

Sample Selection

##Method of Selection

The sample is selected in two stages using `UPsystematic()` from the sampling package (Tille and Matei, 2016), which implements Madow's systematic PPS method. The PSU frame is sorted by FIPS code before selection, which provides implicit geographic stratification — adjacent tracts on the sorted list tend to be geographic neighbors, so the systematic selection spreads the sample across the city (VDK Sec. 3.3). `Set.seed(-77)` was assigned and utilized for reproducibility. Systematic PPS does not permit computation of exact joint selection probabilities ($\pi_{ij} = 0$ for some pairs), which precludes the Horvitz–Thompson variance estimator. In practice, the with-replacement variance approximation via paired analytic strata is used.

###Stage 1: PSU Selection 40 tracts were selected with inclusion probabilities $pi_i = 40 \cdot S_i / S$. The `UPsystematic()` function takes these as input and returns a 0/1 indicator for each PSU. The pi_i range is 0.0136 to 0.3692 with no PSU exceeding 1.0.

###Stage 2: SSU (Block Group) Selection

Within each selected tract, 1 BG was selected with probability proportional to the BG-level composite MOS $\pi_{j|i} = \bar{n} \cdot S_{ij} / S_i$ where $\bar{n} = 1$. For tracts with only a single BG, deterministic selection occurs ($pi_{j|i} = 1$). Overall, 40 BGs were selected with joint selection probability $pi_{ij} = pi_i \cdot pi_{j|i}$ ranges from 0.024 to 0.211.

###Person Selection (Field Procedure, Stage 3)

In area probability sampling, persons are reached through households (VDK Ch. 10). This stage is not executed for this report because no address listing exists, but the procedure is described for field teams to follow in each selected BG. The key outcome variables, vehicle ownership, transit use, skipping trips, asking for rides, are household-level or household-correlated characteristics. If multiple persons from the same household were selected, their responses would be correlated, inflating the design effect and reducing precision. For this reason we recommend selecting one eligible adult per sampled household using a random selection method such as the Kish grid or next-birthday rule. The field procedure within each selected BG is comprised of five

steps: Step 1. Household listing: interviewers compile a complete listing of all occupied housing units using maps and Census address data. Step 2. Household sampling: households are sampled systematically at a rate that, combined with π_{hh} , yields the desired person-level rates. With approximately 1.55 eligible adults per household in Detroit, the HH rate is adjusted accordingly. Step 3. Screening: at each sampled HH, interviewers administer a brief screener to enumerate all persons 18+ and classify each into an age domain. Step 4. Within-HH selection: one eligible adult is selected at random. The selection probability is set to maintain $\pi_{ad} = \pi_{hh} / \pi_{ad}$ for each domain d . Step 5. Verification: the overall probability is $\pi_{ad} \cdot \pi_{hh} \cdot \pi_{ad} = \pi_{ad}$, confirming self-weighting. The expected sample sizes are not integers, as VDK notes after Example 10.1, it is important to sample at the specified rates rather than rounding to fixed counts.

Anticipated Precision

For the key survey variable, the proportion of Detroit residents whose main means of transportation is public transit, precision under the cluster design depends on the design effect. For a two-stage cluster sample, the approximate design effect is (VDK Ch. 9, Eq. 9.5):

$$\text{deff} \approx 1 + (\bar{b} - 1) \delta$$

where $\bar{b}$ is the average number of respondents per PSU (≈ 25 in our design) and δ is the intraclass correlation coefficient measuring homogeneity within tracts. Values of δ for demographic/behavioral variables in U.S. urban populations typically range from 0.01 to 0.05, yielding deffs between 1.2 and 2.2.

The variance of an estimated proportion $\hat{p}$ under the cluster design is approximately:

$$\text{Var}(\hat{p}) \approx \text{deff} \cdot \frac{p(1-p)}{n}$$

An analyst would estimate this using the ultimate cluster variance estimator, which in the with-replacement form is:

$$v_{PSU} = \frac{m}{m-1} \sum_{i \in s} \left(\sum_{k \in s_i} w_k y_k - \frac{\sum_{i \in s} \sum_{k \in s_i} w_k y_k}{m} \right)^2$$

This is implemented automatically by `survey::svymean()` when the design object specifies PSUs and strata (see Sec. 6).

The table below shows the standard error, coefficient of variation, and 95% margin of error for a range of proportions and design effects at $n = 1000$ total respondents.

```
expand_grid(p = c(.10, .15, .20, .25, .30), deff = c(1.5, 2.0, 2.5)) |>
  mutate(n = 1000,
         se = sqrt(deff * p * (1 - p) / n),
         cv = se / p,
         moe = 1.96 * se) |>
  mutate(across(where(is.numeric), ~ round(.x, 4))) |>
  kable(n = Inf)
```

p	deff	n	se	cv	moe
0.10	1.5	1000	0.0116	0.1162	0.0228
0.10	2.0	1000	0.0134	0.1342	0.0263
0.10	2.5	1000	0.0150	0.1500	0.0294
0.15	1.5	1000	0.0138	0.0922	0.0271
0.15	2.0	1000	0.0160	0.1065	0.0313
0.15	2.5	1000	0.0179	0.1190	0.0350
0.20	1.5	1000	0.0155	0.0775	0.0304
0.20	2.0	1000	0.0179	0.0894	0.0351
0.20	2.5	1000	0.0200	0.1000	0.0392
0.25	1.5	1000	0.0168	0.0671	0.0329
0.25	2.0	1000	0.0194	0.0775	0.0380
0.25	2.5	1000	0.0217	0.0866	0.0424
0.30	1.5	1000	0.0177	0.0592	0.0348
0.30	2.0	1000	0.0205	0.0683	0.0402
0.30	2.5	1000	0.0229	0.0764	0.0449

For a transit use proportion of approximately $p = 0.20$ with $deff = 2.0$, the 95% margin of error is ± 0.035 , which is adequate for most policy analyses. Domain-level estimates (250 respondents each) would have approximately double the margin of error, around ± 0.07 .

Weighting Plan

Survey weights are numbers assigned to each respondent that allow us to produce estimates representative of the entire Detroit adult population. Without proper weighting, groups that were oversampled or that had higher response rates would be overrepresented in the estimates.

Base weights

The base weight (also called design weight) for a respondent is the inverse of the overall selection probability. At the BG level, our two-stage selection gives a joint probability of $\pi_{ij} = \pi_i \cdot \pi_{j|i}$, so the BG-level base weight is:

$$d_{0,ij} = \frac{1}{\pi_i \cdot \pi_{j|i}}$$

Under the full three-stage self-weighting design, the person-level probability in domain d would be $\pi_k(d) = f_d$ (VDK p. 283), giving a base weight of $d_0(d) = 1/f_d = N_d/n_d^{\text{select}}$ that is constant within each domain (Lecture Ch. 13, Example 13.3).

QC check (Lecture Slide 20): the sum of base weights across the expected sample in each domain should approximate the domain population total N_d .

Unknown eligibility

During field listing, interviewers may encounter housing units whose eligibility cannot be determined — for example, units that appear occupied but where contact is never made, or addresses that turn out to be vacant or demolished. These cases are classified as unknown eligibility (s_{UNK}). The standard approach is to redistribute the weight of unknowns to cases with known eligibility status within adjustment classes formed from frame variables available for all cases (Lecture Slide 30):

$$a_{1b} = \frac{\sum_{i \in s_b} d_{0i}}{\sum_{i \in s_{b,KN}} d_{0i}}$$

where s_b is all sample cases in class b and $s_{b,KN}$ is the subset whose eligibility is known. The adjusted weight becomes $d_{1i} = a_{1b} \cdot d_{0i}$ for cases in $s_{b,KN}$ and zero for unknowns. Adjustment classes could be formed from tract-level Census demographics (e.g., vacancy rates, housing type).

Non-response adjustment

Among cases known to be eligible, some will decline to participate. The weighting class NR adjustment within class c is (Lecture Slide 48):

$$a_{2c} = \frac{\sum_{i \in s_{c,E}} d_{1i}}{\sum_{i \in s_{c,ER}} d_{1i}}$$

where $s_{c,E}$ is the set of all eligible cases and $s_{c,ER}$ is the subset of eligible respondents in class c . A natural choice for adjustment classes is the four age domains, since response rates are expected to vary substantially (0.35 for ages 18–34 vs. 0.65 for 65+). If the response probability is approximately constant within each domain, this simplifies to $a_{2c} \approx 1/r_d$ (Lecture Slide 50).

Variance Estimation

Because the sample is selected in stages (tracts, then block groups), standard formulas that assume simple random sampling would underestimate the uncertainty of survey estimates. Special variance estimation methods are needed that account for the clustering of persons within tracts. We describe two approaches: Taylor series linearization (our primary recommendation) and jackknife repeated replication (an alternative).

Taylor series linearization (primary method)

For systematic PPS selection, exact joint selection probabilities are not available. The standard practice (VDK Sec. 3.3) is to form **paired analytic strata** by grouping adjacent sample PSUs in the FIPS-sorted selection order. With $m = 40$ sample PSUs, this creates 20 strata, each containing 2 PSUs.

Within this framework, `survey::svydesign()` uses the ultimate cluster variance estimator (Lecture Ch. 9 Slide 65) to estimate standard errors via Taylor series linearization. This treats each PSU as a “cluster” and captures all within-PSU variability — from both the SSU and person stages — through the between-PSU variation.

The analyst would set up the survey design object as follows after data collection. The `id` identifies PSUs (tracts), `strata` identifies the paired analytic strata, and `weights` are the final NR-adjusted weights. The `nest = TRUE` argument tells `svydesign` that PSU IDs restart within each stratum.

Jackknife repeated replication (alternative method)

As an alternative to linearization, **delete-one-PSU jackknife (JKn)** can be used (VDK Ch. 15). This method systematically drops one PSU at a time, re-estimates the statistic from each reduced sample, and computes the variance from the spread of these $m = 40$ replicate estimates. JKn is especially useful for nonlinear estimators (e.g., ratios, quantiles) where the Taylor approximation may be less accurate.

Maps (Assignment 7)

Maps of the Detroit sampling frame and the selected sample were produced for this report. The ggmap versions use Stadia Maps basemap tiles (matching the Assignment 7 style) to provide geographic context — roads, water features, and neighborhood boundaries are visible beneath the choropleth layers. The geom_sf fallback versions render without an API key.

Appendix A

Appendix B

A. Sample PSU and SSU listing

The table below lists all 40 selected BGs with their Census population data, composite MOS, selection probabilities at each stage (π_i and $\pi_{j|i}$), the joint BG probability (π_{ij}), and the domain-specific person sampling rates that field teams would implement. The rates are listed in the `rate_` columns.

INSERT HERE

B. Export files

INSERT HERE

C. AI Disclosure

This project utilized Claude (Anthropic, Opus 4.6) as an AI coding assistant. The tool was used to debug R code throughout development, including diagnosing vector recycling issues with `across()` when multiplying domain columns by the `f_d` vector, resolving `rename()` conflicts caused by duplicate column names after the PSU linking step, and troubleshooting `kable()` and `kableExtra` table formatting for PDF output such as fitting wide tables using `scale_down` and splitting listings into separate display tables. Claude also assisted with map rendering — suppressing verbose `tigris` download output during Quarto rendering and configuring `ggmap` basemap layers with `inherit.aes = FALSE` for proper `geom_sf` overlay alignment. All AI outputs were reviewed, tested, and modified by team members before inclusion.